

KGPL Scan Suite — Full Tile Audit

All 18 tiles · code correctness, scanning, auth/grant gating, live check · 08.09.2026

Scope & method

Every tile was reviewed across four dimensions: **code correctness & logic** (null derefs, stale-response races, OData filter/escaping, field mapping, error handling, double-submit), **scanning / wedge** (auto-submit, focus, camera fallback), and **auth & grant gating** (feature flags vs. backend guards). The single-file app (index.html, 8,914 lines) was read in five parallel passes, then the findings were verified against the source before any change. Ten automated browser test suites were run (all green), and the live production build was checked in Chrome.

Result	High	Medium	Low	Info
Found	0	7	~15	~6
Fixed this session	—	6	1	—
Open (recommended)	0	1*	~14	~6

No High-severity (data-corruption / security) defects were found. * one Medium item (Security Loading carton-read race) is left open deliberately — see notes. The backend re-validates every posted box, so its worst effect is a stale on-screen state, not a bad post.

The short version

The suite is in good shape. The gating is sound, the scan handling is consistent, and the money actions (posts, confirmations, releases) are guarded against double-submit and, in most tiles, against out-of-order replies. The audit turned up no way to post wrong data through the UI on its own. The six fixes applied this session close real gaps that show up under speed or on specific handheld settings; everything still open is either cosmetic, low-probability, or needs a backend detail confirmed first.

Fixed this session (deployed to your repo, transport KSDK906961)

Tile	Aspect	Sev	What was wrong → fix
Dye Pack + Stock	Scanning	Med	Their scan fields were in the wedge <i>redirect</i> list but missing from the <i>burst-detection</i> list (SCAN_FIELDS). On a DataWedge device set to IME output with no Enter suffix, the barcode filled the field but never auto-submitted — reads as "scanning doesn't work". Added dpScan and stkTerm.
Packing List	Logic	Med	Multi-SO queue: removing a chip <i>before</i> the current order left the "current" pointer sliding onto the wrong order (progress and highlight drifted). Now the index is corrected on removal.
Packing List	Logic / UX	Med	On the last queued order the "Next" control was hidden, so the completion path (clear queue, "all done") was unreachable — the queue could only be cleared manually. It now becomes "Finish queue".
Packing List	Race	Med	loadSalesOrder had no stale-response guard; two quick loads (45s each) could let an old reply overwrite the loaded order and blank its scanned boxes. Added a sequence guard.
QC (all four)	Race	Med	findLot had no stale-response guard (its siblings do). Tapping a pending lot then scanning another could set the active lot to the wrong batch, against which UD/Release then posts. Added a sequence guard.
Knit QC	Error handling	Med	A one-time network blip while loading defect codes cached an empty list for the plant for the whole session, permanently showing "no defect codes". Now it caches only a real result and retries on the next scan.

All six are covered by the browser test suite (pick-queue regressions added for the two queue fixes); ten suites pass. Run `npm run deploy` from the repo to push them live.

Open items — recommended, not yet changed

None of these can post bad data on their own; they are ranked by how much they could confuse an operator. Left unchanged this pass either because they need a backend detail confirmed, or to keep the deploy tight around the verified fixes.

Tile	Sev	Issue & recommendation
Security Loading	Med	boxSeq guard is declared but never wired into the carton reads, so overlapping re-reads can briefly show a confirmed carton as pending again. Left open because a naïve guard would break reloadAllBoxes' completion counter — needs a selection-scoped sequence. Backend still flags any re-scan, so no bad post.
HU Movement (Pack)	Med	The deep-entity POST treats any 200 with a body as success without inspecting per-row type 'E'. If Gateway returns 200 while some boxes failed, it can report success. Recommend scanning returned rows for errors — needs the response field names confirmed first.
Job Card	Low	"To Job Card" after batch create is not gated on the JOBCARD grant (the list buttons are); and loadRecipe / loadWorkcenters show a server error as "none on file". Gate the button; show failText on error.
QC (all four)	Low	Per-row "Save" has no double-submit guard (harmless duplicate post); loadChars / mass.loadLot have no lot-change guard. Disable the row button during its post; add a lot guard.
HU Movement (Pack)	Low	Reservation number is normalized differently by scan vs. manual entry, and the reservation item is not validated before submit. Normalize once at submit; require the item.
Location / Add Location	Low	confirmLocation / loadLocations have no seq guard; a timeout shows "Location not found" for a good label; reprint is capped at 500 racks but reports full success; the location field is not focused on entry (matters for IME-only scanners). Use failText for timeouts; focus on entry; warn when the reprint is truncated.
Inventory / Reconciliation	Low	Pending-count read and loadGap have no seq guard; switching sessions quickly can show the previous session's count / gap rows. Add a seq guard as used elsewhere.
Knit QC	Low	postDqc's error path can call enableDqcButtons(null) if another roll is scanned mid-post (TypeError). Guard with "if (dqcRoll)" or block scans while a DQC post is in flight.
Users (Admin)	Low	Screen is gated on the ADMIN <i>feature</i> , not on IsAdmin; no client-side warning before an admin removes their own access (backend blocks last-admin logout). Add an IsAdmin check and a self-edit confirm.
Stock Enquiry	Info	loadGroups is silent if the material-group read fails; confirm the search sends the term as a fragment match (backend contract) rather than exact matnr. Surface a load error.

Backend items (separate ownership)

WIP Batch → New Batch fails at runtime. The RAP action behind it (ZBP_I_WIP_BATCH_MGMT, package ZKGPL_FIORI) is an empty stub on both KSD and KSP — the front-end call, its validations and its double-submit guard are all correct. Close/Reopen on the Batches tab are fine.	Hand to SOLSYNCH to implement the action. No front-end change.
Frequent logouts — session TTL was 12 hours.	Fixed: raised to 30 days in ZCL_Z SOL_APP_AUTH_DPC_EXT, activated on KSD (transport KSDK906959). Applies to new logins; sign out/in once to pick it up.

Per-tile verdict

Tile	Verdict	Notes
Location (LOC)	OK	Read/confirm; two Low race/UX notes open.
Add Location (ADDLOC)	OK	Print-only; reprint cap + seq notes open.
HU Movement (PACK)	Attention	One Med (deep-entity error) + Low open.
Inventory (INV)	OK	One Low session-count race.
Reconciliation (RECON)	OK	One Low gap-load race.
Packing List (PICK)	Fixed	Three Med + one Low queue fixes applied.
Security Loading (DISPATCH)	Attention	One Med carton-read race left open (safe).
Create Challan (CHALLAN)	Clean	Truck + BoxData + guards all correct.
Knit QC (KNITQC)	Fixed	Defect-cache fixed; one Low null-deref open.
WIP Batch (WIPBATCH)	Clean	Front-end fine; New Batch blocked by backend stub.
Job Card (JOBCARD)	OK	Three Low (gating, error masking) open.
Dye Pack (DYEPACK)	Fixed	Scan auto-submit fixed.
Stock Enquiry (STOCK)	Fixed	Scan auto-submit fixed; two Info.
QC Raw (QCRAW)	Fixed	findLot race fixed; Low guards open.
QC Post-Dyeing (QCDYE)	Fixed	Shares the QC engine fix.
QC Post-Winding (QCWIND)	Fixed	Shares the QC engine fix.
QC Inspection (QCINSP)	Fixed	Shares the QC engine fix.
Users (ADMIN)	OK	Low/Info: IsAdmin gate, self-lockout confirm.

KGPL Scan Suite · tile audit · 0 High, 6 Medium fixed · frontend on transport KSDK906961 (run npm run deploy), session-TTL backend on KSDK906959.